

CSCI 101: Quiz2

Problem 1

Introduction:

Our DNA (deoxyribonucleic acid) is the molecule that contains the genetic code of organisms. DNA is composed of four nucleotides, **A, T, C and G**.

We (human-beings) have numerous DNA fragments in our DNA library. In molecular biology, a library is a collection of DNA fragments that is stored and propagated in a population of micro-organisms through the process of molecular cloning.

We want to create a software that automatically distinguish each DNA molecule through attaching a unique DNA barcode at the end of each fragment. DNA barcoding is the method utilized for species identification using a short section of DNA from a specific gene or genes.

However, according to research done in the area, a DNA barcode in needs to be unique (from each other), and must satisfy the following four criteria:

1. Each DNA barcode needs to have a **GC-content from 40% to 60%** (GC-content definition is below)
 - GC-content (or guanine-cytosine content) is the percentage of nitrogenous bases in a DNA or RNA molecule that are either guanine (G) or cytosine (C); The equation is given by:

$$\frac{\text{Count of (G)} + \text{Count of (C)}}{\text{Count of (A)} + \text{Count of (T)} + \text{Count of (G)} + \text{Count of (C)}} \times 100\%$$

2. Any barcode that contains more than three consecutive redundancy (repeated) must be voided. In other words, **there should not be three identical nucleotides in a row**, such as AAA or TTT or CCC or GGG
3. Barcodes **must exclude** restriction sites for AgeI, Ascl, BamHI and SbfI listed below:
AgeI = ACCGGT
Ascl = GGCGCGCC
BamHI = GGATCC
SbfI = CCTGCAGG



Requirement:

Generate a software (Java class name it `DNABarcodeGenerator`) that will do the following:

- Ask the user how many sequences of DNA barcodes (N) she/he wants?
- Ask the user what is the length of the DNA barcode (L) she/he wants?
- And according to the user parameters you must generate a total of **N random sequences** of **L nucleotides** long DNA barcodes and print them to the console one per line

Solution Guidelines:

- Use methods and comment your code
- I expect four methods plus main method (you can add any additional methods you like as long as your code contain the four methods, below)
 - `generateBarcode ()`
 - takes no argument and return a random string of nucleotides of the length, L, which is inputted by the user
 - `isRestrictedList(String barcode)`
 - Takes the generated barcode and validate if the generated barcode is in the restricted list or not, if it is, ignore it and generate a new barcode
 - `validateGCCCount(String barcode)`
 - Takes the generated barcode and validates that count of G nucleotides and the count of C nucleotides is greater or equal than 40% and less or equal than 60% of the total count of nucleotides in the generated barcode
 - `isRedundantBarcode(String barcode)`
 - Takes the generated barcode and validates there are no three nucleotides on a row for given barcode
 - Use class variables when reading the input from the user
 - Assume Barcode length is no less than 4 ... you don't have to validate users' input
 - Assume the count of DNA barcode sequences to generate is no less than 3, you don't have to validate user's input

Example:

- How many sequences of DNA barcodes you would like to generate?
3
- What is the length of the DNA barcode?
8

Output:



barcode1: GTGTACGT
barcode2: CTGACAAC
barcode3: CTGATGAG

Note:

If you follow all the rules here, you may encounter a different result than the ones given in the example since the barcode is a randomly generated sequence

Submittal

- Only submit the java program Birghtspace.
- Add your First & last name and your id as comments in the main method

Rubric

- Code doesn't compile => you will get 0 in the quiz
- Method doesn't provide the correct output => -2pt per method
- output is not formatted as per the instruction => -2pt per each violation
- Missing one of the requirements or not following the guidelines=> -2pt per miss.

This is an individual assignment. Any attempt of cheating will result in getting 0 in the quiz.

Good Luck!
Dr. Nassar